

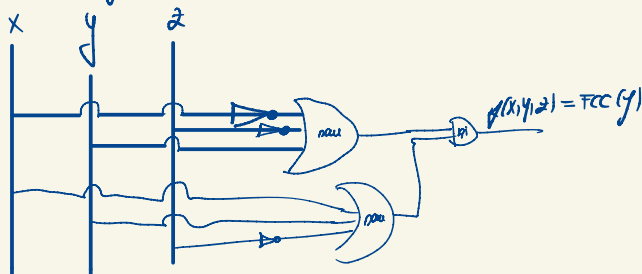

Biletul (1)

x	y	z	$f(x, y, z)$	min term / Max term
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	0	M_2
0	1	1	1	m_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	M_6
1	1	1	1	m_7

$$f(x, y, z) = M_2 \wedge M_6 = (\bar{x} \vee y \vee \bar{z}) \wedge (x \vee y \vee \bar{z}) = FCC(y)$$

$$f(x, y, z) = m_0 \vee m_1 \vee m_3 \vee m_4 \vee m_5 \vee m_7 = \bar{x}\bar{y}\bar{z} \vee \bar{x}\bar{y}z \vee \bar{x}y\bar{z} \vee \bar{x}yz \vee x\bar{y}\bar{z} \vee xy\bar{z} = FCC(y)$$

Circuit FCC(f)



Analiza celălată

Teoria diferență:

Forma canonică conjunctivă, FCC, este conjunctivă maxtermilor corespunzători argumentelor pt. care funcția ia valoarea 0.

Forma canonică disjunctivă, FCD, este disjunctivă mintermilor corespunzători argumentelor pentru care funcția ia valoarea 1.

Fie $f: (B_2)^n \rightarrow B_2$, $n \in \mathbb{N}^*$, o funcție booleană cu n variabile.
 O conjuncție de variabile se numește minterm.

Un minterm care conține toate cele n variabile se numește minterm maxim sau minterm de n variabile. Are forma $x_1^{a_1} \wedge \dots \wedge x_n^{a_n}$, $a_i \in B_2$

Disjuncția care conține toate cele n variabile, având forma:

$x_1^{a_1} \vee \dots \vee x_n^{a_n}$, $a_i \in B_2$ se numește max term de n variabile

Biletul (2)

$$S_f = \{(1, 0, 1, 1), (1, 0, 0, 1), (1, 0, 1, 0), (0, 1, 0, 1), (0, 1, 0, 0), (1, 0, 0, 0), (0, 0, 0, 0)\}$$

	x_1	x_2	x_3	x_4	
<u>I</u>	1	0	1	1	m_{11}
<u>II</u>	1	0	0	1	m_9
<u>III</u>	1	0	1	0	m_{10}
<u>IV</u>	0	1	0	1	m_5
<u>V</u>	0	1	0	0	m_4
<u>VI</u>	1	0	0	0	m_8
<u>VII</u>	0	0	0	0	m_0

$$\underline{V} = \underline{I} + \underline{II} \quad \begin{array}{l} \sqrt{1 \quad 0 \quad - \quad 1} \quad m_{11} \vee m_9 \\ \sqrt{1 \quad 0 \quad 1 \quad -} \quad m_{11} \vee m_{10} \end{array}$$

$$\underline{VI} = \underline{IV} + \underline{III} \quad \begin{array}{l} \sqrt{1 \quad 0 \quad 0 \quad -} \quad m_9 \vee m_8 \\ \sqrt{1 \quad 0 \quad - \quad 0} \quad m_{10} \vee m_8 \end{array}$$

$$\underline{VII} = \underline{III} + \underline{IV} \quad \begin{array}{l} 0 \quad - \quad 0 \quad 0 \\ - \quad 0 \quad 0 \quad 0 \end{array}$$

$$\underline{VIII} = \underline{V} + \underline{VI} \quad \begin{array}{l} 1 \quad 0 \quad - \quad - \\ \hline \hline \hline \end{array}$$

Grupăm mintermii după numărul de variabile descrescătoare

Construim prima tabelă, fiecare grup conținând mintermii cu același nr. de cifre 1. Tragem linie dublă.

Începem factorizarea simplă. Factorizăm doar grupuri adiacente. Mintermii care factorizează trebuie să difere printr-un singur minterm.

Acord simplă une diferă este în locuri cu -1

Bifurcăm mintermii care factorizează. Ei nu sunt mintermii maxime.

$$m_{11} \vee m_9 = \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 = \bar{x}_1 \bar{x}_2 \bar{x}_3 = m_{max1}$$

$$m_9 \vee m_8 = \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 = \bar{x}_1 \bar{x}_2 \bar{x}_3 = m_{max2}$$

$$m_8 \vee m_0 = x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 = \bar{x}_2 \bar{x}_3 \bar{x}_4 = m_{max3}$$

Factorizare dublă
 $m_{11} \vee m_9 \vee m_{10} \vee m_8 =$

$$= x_1 \bar{x}_2 \bar{x}_3 x_4 \vee x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 \bar{x}_3 x_4 \vee x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 = x_1 \bar{x}_2 \bar{x}_3 = m_{max4}$$

$$M(y) = \{ \underline{x_1 x_2 x_3}, \overline{x_1} \overline{x_2} \overline{x_3}, \overline{x_2} \overline{x_3} \overline{x_4}, \underline{x_1 x_2} \}$$

	$\max_1$	$\max_2$	$\max_3$	$\max_4$
m_5	*			
m_4	*	*		
m_0		*	*	
m_8			*	*
m_{11}				*
m_9				*
m_{10}				*

$$C(y) = \{a_{max}, a_{max}\}$$

Cazul 4

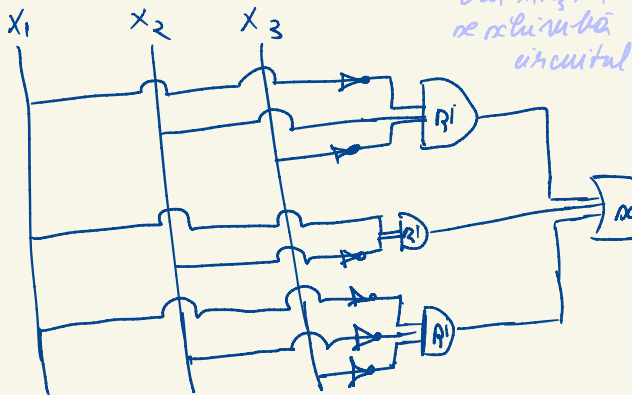
$$g(x_1, x_2, x_3, x_4) = \max_1 \vee \max_2 = \overline{x_1} \vee x_2 \vee \overline{x_3} \vee x_1 \vee \overline{x_2}$$

Singulare minimă în număr de perechi este cu

$$h_1(x_1, x_2, x_3, x_4) = \max_2$$

$$h_2(x_1, x_2, x_3, x_4) = \max 3$$

$$f'(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) \vee \overline{h_1}(x_1, x_2, x_3, x_4) = \overline{x_1} x_2 \overline{x_3} \vee x_1 \overline{x_2} \vee \overline{x_1} x_2 \overline{x_3}$$



inceperu factorizarea
dubla. Se optica acetasi
reguli.

Nullfinesse monese melior
maximale este formată
din monedele nebijate

Construim a doua tabelă
Pe axii: nu interzicem ca
fie parte din monocornule
maximale.

Решение: Попробуем
maximize

*: un'intervall face
parte din monom

⊗: Stetide n'ingure pe linie

Hoguram' coloa' ule case
contin (a). Ele sunt morma-
'ule continue.

Hapwram liwile ce
contîn & hapwate

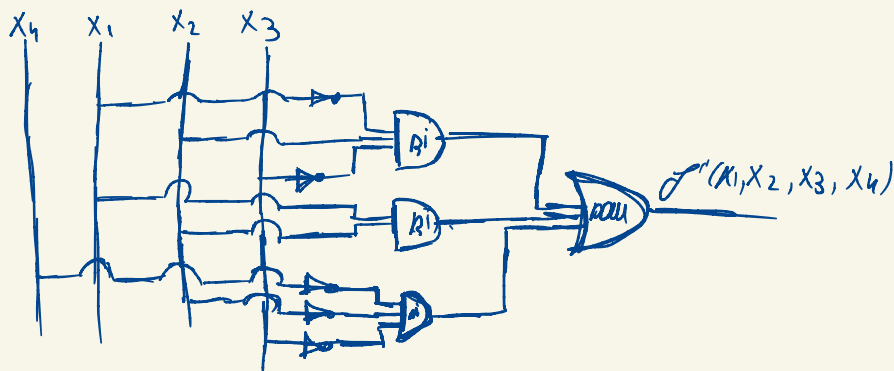
Voir avec deux fonctions
 h , augmentante maximale
et g continue

(Imo este născuț
mi interm al lui
care mi apare în g)

Avec deux monomômes
modulaires ce il
contient $p \equiv 0 \Rightarrow$
" =, donc forme
 (x_1, x_2, x_3, x_4) simplifi-
cote

$$f'(x_1, x_2, x_3, x_4)$$

$$f''(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) \vee h_2(x_1, x_2, x_3, x_4) \\ = \overline{x_1} x_2 \overline{x_3} \vee x_1 \overline{x_2} \vee \overline{x_2} x_3 \overline{x_4}$$



Biletul (3)

$$f(x_1, x_2, x_3, x_4) = \overline{x_1} \overline{x_2} \overline{x_3} x_4 \vee x_1 x_2 x_3 x_4 \vee x_1 \overline{x_2} x_3 \overline{x_4} \vee \overline{x_1} \overline{x_2} x_3 x_4 \vee x_1 x_2 \overline{x_3} \overline{x_4} \vee \overline{x_1} \overline{x_2} \overline{x_3} \overline{x_4} \vee x_1 \overline{x_2} x_3 x_4 \vee x_1 x_2 x_3 \overline{x_4} \vee \overline{x_1} \overline{x_2} x_3 \overline{x_4}$$

$x_1 x_2 \backslash x_3 x_4$	00	01	11	10
00	m_0	m_1	m_3	m_2
01				
11	m_{12}		m_{15}	m_{14}
10			m_{11}	m_{10}

Introducere în funcție sum-
termii

Construirea diagrama

Karnaugh pentru 4
variabile

Planul quintorului pe
pozițiile corecte pentru toate

$$max_1 = m_0 \vee m_1 \vee m_3 \vee m_2 = \overline{x_1} \overline{x_2}$$

$$max_2 = m_5 \vee m_{14} \vee m_{11} \vee m_{10} = x_1 x_3$$

$$max_3 = m_{11} \vee m_{10} \vee m_3 \vee m_2 = \overline{x_2} x_3$$

$$max_4 = m_{14} \vee m_{12} = x_1 x_2 \overline{x_4}$$

$$H(f) = \{max_1, max_2, max_3, max_4\}$$

$$C(f) = \{max_1, max_2, max_4\}$$

$$\left. \begin{array}{l} H(f) \rightarrow C(f) \\ C(f) \rightarrow \emptyset \end{array} \right\} \Rightarrow \text{tabelul de simplificare}$$

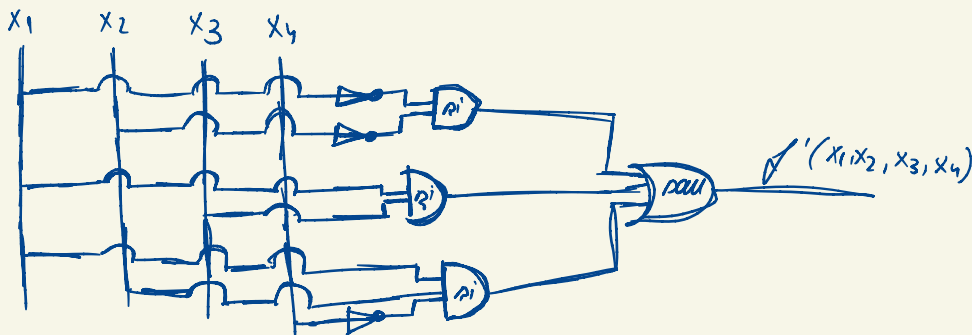
Monomiale maximele sunt
grupurile de câte 2^k de min-
termi adiacenți, neinter-
sectați de câte grupuri

Multiplu monomialelor centra-
le este formată din acele
monomiale care au cel puțin
un quintor care le
aparține doar lor

$g(x_1, x_2, x_3, x_4) = \max_1 \vee \max_2 \vee \max_3$
 Nu mă uităm nimic la necoresp.

g. disjunctivă monotonă met. centrale
 Toti minitermii funcției g apar sing.

$$\Rightarrow f'(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) = \overline{x_1} \overline{x_2} \vee x_1 x_3 \vee x_1 x_2 \overline{x_4}$$



Biletul 15)

$$f(x_1, x_2, x_3, x_4) = x_1 \overline{x_2} \overline{x_3} x_4 \vee x_1 x_2 x_3 x_4 \vee x_1 \overline{x_2} x_3 \overline{x_4} \vee x_1 x_2 \overline{x_3} x_4 \vee \overline{x_1} \overline{x_2} x_3 x_4 \vee x_1 x_1 \overline{x_3} \overline{x_4} \vee \overline{x_1} \overline{x_2} \overline{x_3} x_4 \vee x_1 \overline{x_2} x_3 x_4 \vee x_1 x_2 x_3 \overline{x_4} \vee \overline{x_1} \overline{x_2} x_3 \overline{x_4}$$

	x_1	$\overline{x_1}$		
x_2	m_{15}	m_{13}		x_4
	m_{14}	m_{12}		$\overline{x_4}$
$\overline{x_2}$	m_{10}		m_0	m_2
	m_{11}	m_9		x_4
	x_3	$\overline{x_3}$	x_3	

construim diagonala lui Vatch

Puneam minitermii pe poz. coresp.

Grupăm minitermii în grupuri de câte 2^k , grupurile au fiind numărate de câte grupuri

Disjunctivă minitermilor din aceste grupuri formăm 25 monomii normale.

$$\max_1 = m_{15} \vee m_{13} \vee m_{14} \vee m_{12} = x_1 x_2$$

$$\max_2 = m_{15} \vee m_{13} \vee m_{11} \vee m_9 = x_1 x_4$$

$$\max_3 = m_2 \vee m_3 \vee m_{10} \vee m_{11} = \overline{x_2} x_3$$

$$\max_4 = m_0 \vee m_2 = \overline{x_1} \overline{x_2} \overline{x_3}$$

$$+ \max_5 = m_{15} \vee m_{14} \vee m_{10} \vee m_{11}$$

Monomii centrale sunt acele care au cel puțin un monom pe care nu îl are niciun alt grup.

$$M(f) = \{m_{\max 1}, m_{\max 2}, m_{\max 3}, m_{\max 4}, m_{\max 5}\} \Rightarrow \text{Scriem în notă canonică II}$$

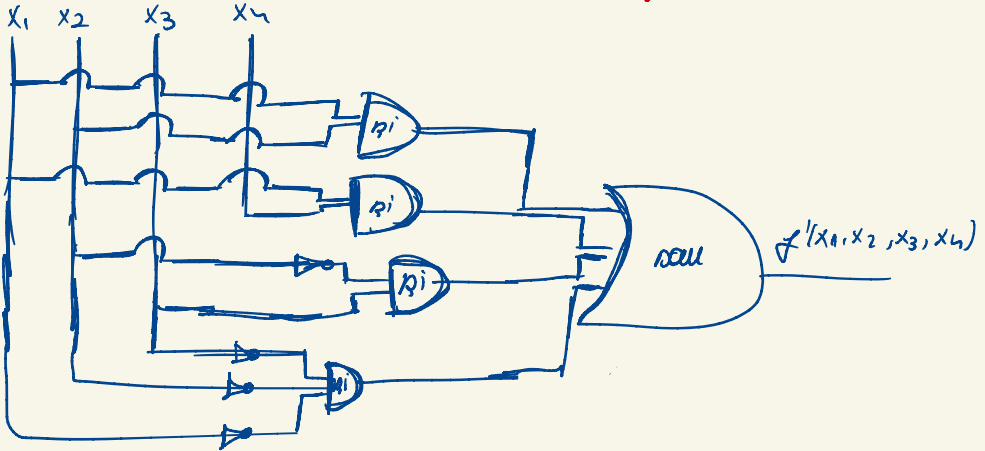
$$C(f) = \{m_{\max 1}, m_{\max 2}, m_{\max 3}, m_{\max 4}\}$$

$$g(x) = m_{\max 1} \vee m_{\max 2} \vee m_{\max 3} \vee m_{\max 4}$$

$$= x_1 x_2 \vee x_1 x_4 \vee \overline{x_2} x_3 \vee \overline{x_1} \overline{x_2} \overline{x_3}$$

Nu mai rămâne nimic de simplificat $\Rightarrow$

$$f'(x_1, x_2, x_3, x_4) = m_{\max 1} \vee m_{\max 2} \vee m_{\max 3} \vee m_{\max 4} = g(x) = x_1 x_2 \vee x_1 x_4 \vee \overline{x_2} x_3 \vee \overline{x_1} \overline{x_2} \overline{x_3}$$



Biletul (4)

$$f(x, y, z) = m_0 \vee m_1 \vee m_2 \vee m_3 \vee m_5 \vee m_6 = FCD$$

$$\rightarrow S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1), (0, 1, 0), (0, 0, 1), (0, 0, 0)\} = \text{nu sîi amze}$$

supra a unității mi-

lor

	x_1	x_2	x_3	
$\overline{I}$	$\sqrt{1}$	1	0	m_6
	$\sqrt{1}$	0	1	m_5
	$\sqrt{0}$	1	1	m_3
$\overline{II}$	$\sqrt{0}$	1	0	m_2
	$\sqrt{0}$	0	1	m_1
$\overline{III}$	$\sqrt{0}$	0	0	m_0

$\overline{IV} = \overline{I} + \overline{II}$	—	1	0
	—	0	1
	$\sqrt{0}$	1	—
	$\sqrt{0}$	—	1
$\overline{V} = \overline{II} + \overline{III}$	$\sqrt{0}$	—	0
	$\sqrt{0}$	0	—

Factorizare simplă

$$m_6 \vee m_2 = m_{\max 1} = y \overline{z}$$

$$m_5 \vee m_1 = m_{\max 2} = y \overline{x}$$

$$m_3 \vee m_2$$

$$m_3 \vee m_1$$

$$m_2 \vee m_0$$

$$m_1 \vee m_0$$

Factorizare dublă

101
001

$$\overline{m_1} = \overline{m_2} \vee \overline{m_3} \vee \overline{m_4} \vee \overline{m_5} \vee \overline{m_6} \vee \overline{m_7} = X$$

$$H(y) = \{m_{\max 1}, m_{\max 2}, m_{\max 3}\}$$

	$m_{\max 1}$	$m_{\max 2}$	$m_{\max 3}$
m_6	$\textcircled{*}$		
m_2	$*$		$*$
m_5		$\textcircled{*}$	
m_1		$*$	$*$
m_3			$\textcircled{*}$
m_0			$\textcircled{*}$

$$C(y) = \{m_{\max 1}, m_{\max 2}, m_{\max 3}\} \Rightarrow \text{cazuri 1}$$

$$f'(x_1, y_1, z) = y_1 \bar{z} \vee \bar{y}_1 z \vee x_1$$

+ circuitul la f' in z

Biletul 6)

$$f(1,0,0) = f(0,1,1) = 0 \rightarrow f(0,0,0) = f(0,0,1) = f(0,1,0) = f(1,0,1) = f(1,1,0) =$$

$$= f(1,1,1) = 1$$

$$\Rightarrow f(x_1, x_2, x_3) = m_0 \vee m_1 \vee m_2 \vee m_5 \vee m_6 \vee m_7 = FCB =$$

	x_1	$\bar{x}_1$	
x_2	m_6	m_7	m_2
$\bar{x}_2$		m_5	m_1
	$\bar{x}_3$	x_3	$\bar{x}_3$

$$= \bar{x}_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3 \vee$$

$$\vee \bar{x}_1 x_2 \bar{x}_3 \vee x_1 \bar{x}_2 x_3 \vee$$

$$\vee x_1 x_2 \bar{x}_3 \vee x_1 x_2 x_3$$

$$m_{\max 1} = m_6 \vee m_2 = x_2 \bar{x}_3$$

$$m_{\max 2} = m_6 \vee m_7 = x_1 x_2$$

$$m_{\max 3} = m_7 \vee m_5 = x_1 x_3$$

$$m_{\max 4} = m_5 \vee m_1 = \bar{x}_2 x_3$$

$$m_{\max 5} = m_1 \vee m_0 = \bar{x}_1 \bar{x}_2$$

$$m_{\max 6} = m_0 \vee m_2 = \bar{x}_1 \bar{x}_3$$

$$M(f) = \{\max_1, \dots, \max_7\} \quad \Rightarrow \text{Complexität}$$

$$C(f) = \emptyset$$

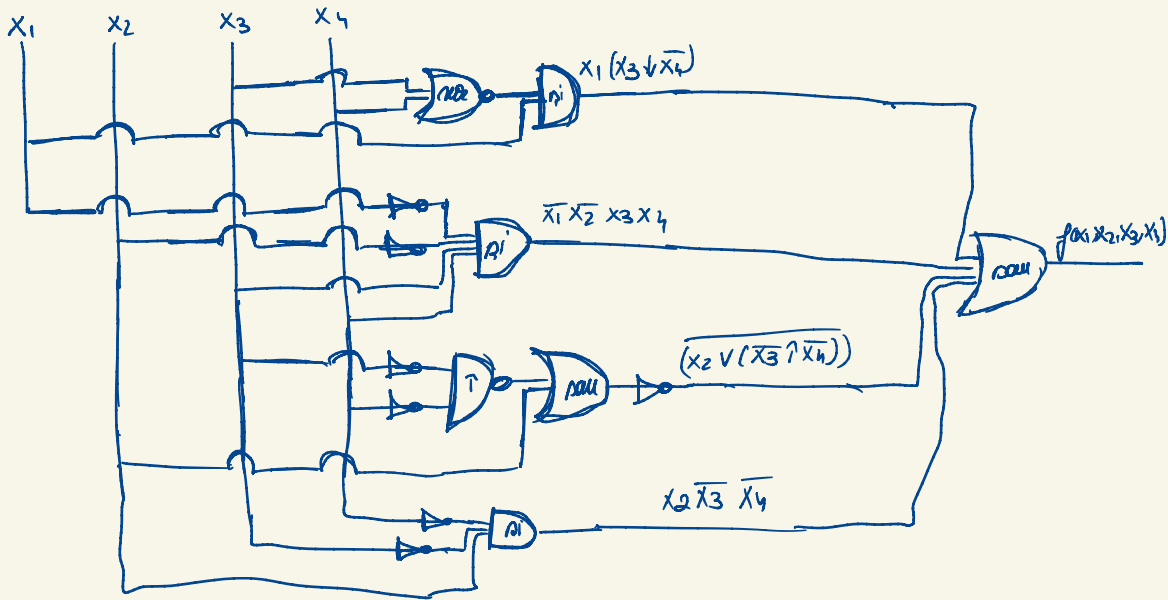
$$f'(x_1, x_2, x_3, x_4) = \max_1 \vee \max_3 \vee \max_5 = x_2 \bar{x}_3 \vee x_1 x_3 \vee \bar{x}_1 \bar{x}_2$$

$$f''(x_1, x_2, x_3, x_4) = \max_2 \vee \max_4 \vee \max_6 = x_1 x_2 \vee \bar{x}_2 x_3 \vee \bar{x}_1 \bar{x}_3$$

→ Circuitele

Biletul 7)

$$f(x_1, x_2, x_3, x_4) = x_1(x_3 \downarrow \bar{x}_4) \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee (x_2 \vee (\bar{x}_3 \uparrow \bar{x}_4)) \vee x_2 \bar{x}_3 \bar{x}_4$$



$$f(x_1, x_2, x_3, x_4) = x_1(x_3 \downarrow \bar{x}_4) \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee (x_2 \vee (\bar{x}_3 \uparrow \bar{x}_4)) \vee x_2 \bar{x}_3 \bar{x}_4$$

$$= x_1(\overline{x_3 \vee \bar{x}_4}) \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee (x_2 \vee (\bar{x}_3 \wedge \bar{x}_4)) \vee x_2 \bar{x}_3 \bar{x}_4$$

$$= x_1 \bar{x}_3 x_4 \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee (x_2 \vee x_3 \bar{x}_4) \vee x_2 \bar{x}_3 \bar{x}_4$$

$$= x_1 \bar{x}_3 x_4 \vee \bar{x}_1 \bar{x}_2 x_3 x_4 \vee x_2 \bar{x}_3 \bar{x}_4 \vee x_2 x_3 \bar{x}_4$$

$$x_1 \bar{x}_3 x_4 = x_1(x_2 \vee \bar{x}_2) \bar{x}_3 x_4 = x_1 x_2 \bar{x}_3 x_4 \vee x_1 \bar{x}_2 \bar{x}_3 x_4$$

$$\bar{x}_2 \bar{x}_3 \bar{x}_4 = (\bar{x}_1 \vee x_1) \bar{x}_2 \bar{x}_3 \bar{x}_4 = x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4$$

$$x_2 \bar{x}_3 \bar{x}_4 = (x_1 \vee \bar{x}_1) x_2 \bar{x}_3 \bar{x}_4 = x_1 x_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4$$

$$\rightarrow f(x_1, x_2, x_3, x_4) = \overset{m_0}{x_1 x_2 \bar{x}_3 \bar{x}_4} \vee \overset{m_4}{x_1 \bar{x}_2 \bar{x}_3 x_4} \vee \overset{m_3}{x_1 \bar{x}_2 x_3 x_4} \vee \overset{m_7}{x_1 \bar{x}_2 x_3 \bar{x}_4} \vee \overset{m_{13}}{\bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4} \vee \overset{m_{12}}{x_1 x_2 \bar{x}_3 \bar{x}_4} \vee \overset{m_5}{\bar{x}_1 x_2 \bar{x}_3 \bar{x}_4} = \Sigma \Delta$$

$x_3 x_4 \backslash x_1 x_2$	00	01	11	10
00	m_0		m_3	
01	m_4			
11	m_{12}	m_{13}		
10	m_8	m_9		

$$m_{\max 1} = m_0 \vee m_4 \vee m_{12} \vee m_8 = \bar{x}_3 \bar{x}_4$$

$$m_{\max 2} = m_8 \vee m_9 \vee m_{12} \vee m_{13} = x_1 \bar{x}_3$$

$$m_{\max 3} = m_3 = \bar{x}_1 \bar{x}_2 x_3 x_4$$

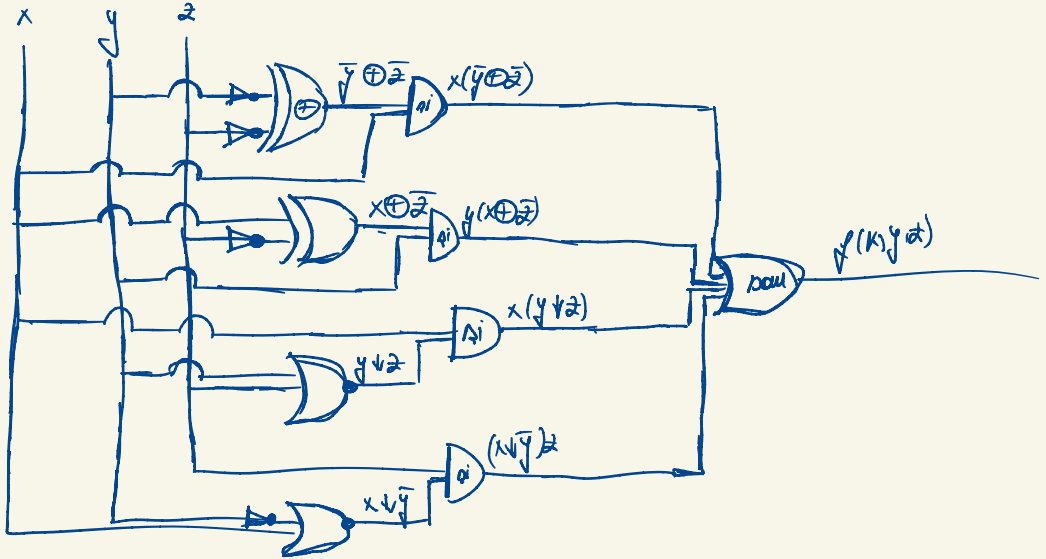
$$\begin{aligned} M(f) &= \{m_{\max 1}, m_{\max 2}, m_{\max 3}\} \\ C(f) &= \{m_{\max 1}, m_{\max 2}, m_{\max 3}\} \end{aligned} \quad \left. \vphantom{\begin{aligned} M(f) \\ C(f) \end{aligned}} \right\} \rightarrow \text{canon. 1}$$

$$\Rightarrow f'(x_1, x_2, x_3, x_4) = m_{\max 1} \vee m_{\max 2} \vee m_{\max 3} \vee m_{\max 4} = \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3 x_4$$

+ DeMorgan

Bibel wapp

$$f(x, y, z) = x(\bar{y} \oplus \bar{z}) \vee y(x \oplus \bar{z}) \vee x(y \vee z) \vee (x \vee \bar{y})z$$



$$x \oplus y = \bar{x}y \vee x\bar{y}$$

$$x(\bar{y} \oplus \bar{z}) = x(\bar{y}\bar{z} \vee yz) = x\bar{y}\bar{z} \vee xy\bar{z}$$

$$y(x \oplus \bar{z}) = y(x\bar{z} \vee xz) = x\bar{y}\bar{z} \vee xy\bar{z}$$

$$x(y \vee z) = x(\overline{y \vee z}) = x\bar{y}\bar{z}$$

$$(x \vee \bar{y})z = (x \vee \bar{y})z = \bar{x}y\bar{z}$$

$$\rightarrow f(x, y, z) = \underbrace{xy\bar{z}}_{m_6} \vee \underbrace{x\bar{y}z}_{m_5} \vee \underbrace{\bar{x}y\bar{z}}_{m_2} \vee \underbrace{xy\bar{z}}_{m_7} \vee \underbrace{x\bar{y}z}_{m_4} = \text{FCB}$$

$\begin{matrix} x \backslash yz \\ yz \end{matrix}$	00	01	11	10
0				m_2
1	m_4	m_5	m_7	m_6

$$m_{\max 1} = m_4 \vee m_5 \vee m_7 \vee m_6 = x$$

$$m_{\max 2} = m_2 \vee m_6 = y\bar{z}$$

$$\begin{aligned} M(y) &= \{m_{\max 1}, m_{\max 2}\} \\ C(y) &= \{m_{\max 1}, m_{\max 2}\} \end{aligned} \quad \left. \vphantom{\begin{aligned} M(y) \\ C(y) \end{aligned}} \right\} \rightarrow \text{Case 1}$$

$$\Rightarrow f'(x, y, z) = m_{\max 1} \vee m_{\max 2} = x \vee y\bar{z}$$

+ Derive